MIL-M-46336C(MU)
AMENDMENT 10
7 October 1970
SUPERSEDING
AMENDMENT 9
11 May 1970

MILITARY SPECIFICATION

MODULES: REPRODUCER, SIGNAL DATA: AN/GSQ-64

This amendment forms a part of Military Specification MIL-M-46336(MU), dated 10 March 1965.

Paragraph 1.1: Delete "Transistor Assembly, Power: 10531682". Add: "Terminal Board TB-6: 10521591".

Paragraph 2.1, under DRAWINGS: Delete "F10531682 Transistor Assembly, Power and F10531685 Schematic, Electrical Transistor Assembly, Power". Add: "D10531591 Terminal Board TB-6 and D10531594 Schematic, Terminal Board TB-6".

Paragraph 2.1, under PACKAGING DATA SHEETS: Delete in entirety and substitute:

"8213319	Packaging of Amplifier Assembly, Photodiode
10531591	Packaging of Terminal Board TB-6
10531686	Packaging of Amplifier Power Supply
10531690	Packaging of Amplifier Electronic Control
10531698	Packaging of Signal Data Reproducer Subassembly".

Paragraph 3.5.2: Change "15g's" to "5g's".

Delete paragraphs 3.6, 3.6.1, 3.6.2 and 3.6.3 in their entirety.

Paragraph 3.7.1(b): Delete in its entirety and substitute the following: The forward voltage drop across the CR5 and CR6 combination shall be between 1.2 and 2.2 volts DC when conducting 100 ± 10MA.

Paragraph 3.7.1(c): Change "9.5K ohms and 11K ohms" to "8.0K ohms and 12.0K ohms" respectively.

Paragraph 3.7.1(d): Delete in its entirety.

Paragraph 3.7.1(f): Change "200K ohms" to "150K ohms".

Paragraph 3.7.4(a): Change "+24.0V ± 1.5V" to "+26.0V ± 2.0V".

Paragraph 3.7.5: Delete in its entirety.

Paragraph 3.7.7: Change "36 ohm load" to "10 ohm load".

Paragraph 3.7.9.1(b): Same as 3.7.1(b).

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* Paragraph 3.7.9.1(c): Change "9.5K" to "8K".

Paragraph 3.7.9.3: Delete and substitute the following: The requirement of 3.7.3 shall apply except at -40°F, the output voltage shall be +5.6 ± 0.5 and at +125°F, +6.1 ± 0.5.

Paragraph 3.7.9.5: Delete in its entirety.

Paragraph 3.8.1.2: Change "Plus 3.0 minus 1.5 microfarads" to "plus 4.0 minus 2 microfarads". Change "51K ohms ± 5%" to "51K ohms ± 8%".

Paragraph 3.8.2: Delete "10 ± 1 MA and substitute "5 MA". Delete "470 ohms ± 5%" and substitute "470 ohms ± 8%.

Paragraph 3.8.3: Delete "91K ohms ± 5%" and substitute "91K ohms ± 8%".

Paragraph 3.8.5(a): Change "-5.75 ± 0.4 volts" to "-5.42 ± 0.4 volts".

Paragraph 3.8.7, Figure 1: Change "+5.9 ± .4" to "0.0 ± 0.3 volts, with a 20K ohm load".

Paragraph 3.8.10: Change "an input of ground" to "an input of +2.0 ± 0.5 volts".

Paragraph 3.8.11.3(a): Delete "-5.7 ± 0.5 volts" and substitute "-5.4 ± 0.5 volts".

Paragraph 3.8.11.8(a): Delete in its entirety and substitute "A plus 2.0 ± 0.2 volts input shall produce an emitter voltage between 0 and minus 0.6 volt".

Paragraph 3.9.1, Figure 2(B): Delete "1.54 ± .1 msec" and substitute "1.54 ± $\frac{1}{3}$ msec"; delete "-11.5 ± 1.5v" and substitute "-9.1 ± 1.8v".

Paragraph 3.9.4: Change "480 ± 20 cps" to "488 ± 20 cps".

Paragraph 3.10: Delete "electrical schematic" and substitute "wiring diagram".

Paragraph 3.10.1: Delete in its entirety and substitute a new paragraph:

"3.10.1 Continuity.- The signal data reproducer subassembly shall exhibit continuity in accordance with Wiring Diagram F10531712."

Paragraph 3.10.2: Change "15.2 to 16.8" to "19.0 to 21.0" and "5 amp" to "125 MA".

Paragraph 3.10.3: Delete first sentence and substitute the following: "Q2 shall have maximum collector'base cutoff current of 15 ma when V_{CBO} - 60V".

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Paragraph 3.10.4: Delete first sentence and substitute the following: "Q1 shall have a maximum collector'base cutoff current of 15 ma when V_{CBO} - 60V."

* Paragraph 3.10.7: Change "0.3 volts" to "0.75 volts".

Paragraph 3.10.8.1: Change "-14.4V and -17.2V" to "-18.0V and -21.7V" and "0.5 Amp" to "125MA".

* Paragraph 3.10.8.6: Change ".5V" to "0.75 volts".

Add new paragraph 3.11:

"3.11 Terminal Board TB-6: 10531591.- The circuitry shall be in accordance with electrical schematic, Drawing D10531594. Unless otherwise specified, the following requirements apply at standard room temperature (60° to 90°F).

Diode.

(a) Forward voltage drop.- When conducting 10ma, diodes CR1 through CR9 shall exhibit not more than 1.0 volt drop. At low temperature (-40°F), the drop shall be not more than 1.5 volts."

(b) Diode reverse current.- With 5 VDC applied in the reverse direction, not more than 5 ua shall flow through the diodes CR1 through CR9. At high temperature (+125°F), the reverse current shall be not more than 35 ua."

Following paragraph 4.1.1, add paragraphs 4.1.2, 4.1.2.1 and 4.1.2.2.

"4.1.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) of each type module manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.1.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.1.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection."

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Paragraph 4.2.3: Add the following sentence: "No lots shall be accepted until control samples have been completely tested."

TABLE I: Delete and substitute:

"TABLE I

Terminal Board TB-6: 10531591
(100 percent inspection)

	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
101.	Diode-forward voltage drop	3.11(a)	4.5.2.6(a), 4.5.2.6(b),
102.	Diode-reverse current	3.11(b)	4.5.2.6(c)

TABLE II, line 5: Delete in its entirety.

Paragraph 4.3.2: Delete first sentence and substitute: "Modules shall be selected at random by a Government representative as control samples from each three (3) months production or portion thereof as follows:

3 units per first 100 of each type produced
5 units per 200 " " "
7 units per 300 " " "
9 units per 400 " " "
10 units per 500 " " "
12 units per 700 " " "
15 units per 1000 " " "
and 2 units per every 100 of each type produced in excess of 1000."

TABLE VII: Delete last column Test Procedures and substitute the following new column:

"Test Procedures

4.5.1.1
4.5.1.2
4.5.1.3.1
4.5.1.3.2"

TABLE VIII: Delete and substitute:

"TABLE VIII

Terminal Board TB-6: 10531591
(-40°F and +125°F)

	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
308.	Diode-forward voltage drop	3.11(a)	4.5.2.6(d)
309.	Diode-reverse current	3.11(b)	4.5.2.6(e)"

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TABLE IX, line 5: Delete in its entirety.

Paragraph 4.4.1.1, Items 4,5,6,7 and 8 under description: Delete "specified in 4.4.2.2" and substitute "specified in 4.4.1".

Paragraph 4.5.1.3.1: Delete last sentence and substitute the following: "The unit shall then be thoroughly dried and then subjected to the test procedures listed in TABLE II for "P/N 10531686; TABLE III for P/N 10531690, TABLE IV for P/N 8213319, and TABLE V for P/N 10531698".

Paragraph 4.5.1.3.2: Delete the last sentence and substitute the following: "Throughout the test period, the board shall be capable of performing the tests of TABLES II, III, IV and V as applicable for each module. In order to demonstrate this capability, the tests in TABLES II, III, IV and V shall be performed a minimum of once each 6 hours of the 24 hour test period. The final test shall be conducted during the final portion of the 24 hour period. Between tests, the boards shall be electrically de-energized."

Delete the following in their entirety: Paragraphs "4.5.2.1, 4.5.2.1.1, 4.5.2.1.2, 4.5.2.1.3, 4.5.2.1.4, 4.5.2.1.4.1, 4.5.2.1.4.2, Figures 4 and 5, TABLES XIII, XIV and TABLE XV.

Paragraph 4.5.2.2.1.1: Add new paragraph

"4.5.2.2.1.1 Apply -12 ± 0.1 VDC through a 100 ohm ± 1% 1/4 watt resistor to Pin 3 and ground Pin 8. The voltage between Pin 3 and Pin 8 shall be between 1.2 and 2.2 VDC."

TABLE XVI, third column: Change "9500" to "9000". Change "200K" to "150K"; Fourth column: Change "11,000" to "12,000"; Last column: Delete last seven lines and substitute the following:

"3.7.1(e)
3.7.1(f)
3.7.1(f)
3.7.1(f)
3.7.1(f)
3.7.1(f)
3.7.1(f)";

Delete lines 13 and 17.

Paragraph 4.5.2.2.3(a): Delete "+6 ± 0.5 volts" and substitute "5.9 ± 2.0 volts".

Paragraph 4.5.2.2.4(a): Change "+24.0 ± 1.5 volts" to "+26.0 ± 2.0 volts".

Paragraph 4.5.2.2.5: Delete in its entirety.

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Paragraph 4.5.2.2.7(a), after the second sentence add: "Connect a 10K ohm ± 5% 25 watt resistor between Pin 11 and -150 volt supply.

Paragraph 4.5.2.2.7(b): Delete first sentence; Delete "5.7 volts" and substitute "5.5 volts"; Change "36 ohm" to"10 ohm".

Paragraph 4.5.2.2.7(c): Change "36 ohm" to "10 ohm".

Paragraph 4.5.2.2.8(a)(4): Delete "-7.1 ± 0.4 volts" and substitute "-7.1 ± 0.9 volts".

Paragraph 4.5.2.2.8(b)(2): Delete "shall be -15 ± 1.5 volts" and substitute "with respect to -15 volts shall be less than 0.16 volt".

Paragraph 4.5.2.2.8(c), after the second sentence add: "Remove ground from Pin 39 and apply -12.5 volts".

Paragraph 4.5.2.3.1.2: Delete "of 51K ohms ± 5%" and substitute "specified in 3.8.1.2". Change "plus 3.0 minus 1.5 microfarads" to "plus 4.0 minus 2.0 microfarads".

Paragraph 4.5.2.3.2(a): Delete "of 470 ohms ± 5%" and substitute "specified in 3.8.2".

Paragraph 4.5.2.3.2(b): Delete in its entirety and substitute the following: "With 6.0 ± 1.0 VDC applied to pin 60 and ground to pin 25, adjust the voltage to obtain 5 MA through R42. The voltage at pin 35 shall be 3.5 ± 0.5 VDC to comply with 3.8.2.".

Paragraph 4.5.2.3.3: Delete "of 91K ohms ± 5% to determine compliance with 3.8.3" and substitute "specified in 3.8.3".

Paragraph 4.5.2.3.4, line 2: Change "Pin 24" to "Pin 34".

Paragraph 4.5.2.3.5.1: Delete "-5.75 ± 0.4 volts" and substitute "-5.42 ± 0.5 volts".

Paragraph 4.5.2.3.7: Delete second sentence and substitute the following: "Connect a 20K ohm ± 1% load resistor from Pin 12 to ground. Connect an oscilloscope across the load".

Paragraph 4.5.2.3.8, last sentence: Change "0.2V ± 0.02V" to "-0.2V ± 0.2V".

Paragraph 4.5.2.3.10(b): Change "with ground" to "with +2.0 ± 0.5 volts".

Paragraph 4.5.2.4.5, last sentence: Change "Pin 4" to "Pin 2".

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Paragraph 4.5.2.5.1: Delete in its entirety and substitute the following:

"4.5.2.5.1 Continuity.- The point to point wiring shall be tested with a conventional type tester for compliance with 3.10.1 prior to performing the tests specified in 4.5.2 .5.2 through 4.5.2.5.8.6."

Paragraph 4.5.2.5.2: Delete in its entirety and substitute the following:

"4.5.2.5.2 Zener voltages.- To determine compliance with 3.10.2, apply -30 volts to J9, Pin 3. Place a 75 ohm, 2 watt, 5% resistor between J9, Pin 9 and J9, Pin 10. Place ground at J9, Pin 10, Read -19.0 to -21.0 volts at J9, Pin 9."

Paragraph 4.5.2.5.3(a): Delete in its entirety and substitute the following:

"(a) Apply -100 volts directly to J3, Pin 17. Place ground at J3, Pin 24. Through a microammeter (Item 4, paragraph 4.4.1.1), read a maximum current of 15ma."

Paragraph 4.5.2.5.3(b): Delete in its entirety.

Paragraph 4.5.2.5.4(a): Delete in its entirety and substitute the following:

"(a) Apply -60 volts directly to J2, Pin 65. Place ground at J2, Pin 64. Through a microammeter (Item 4 of 4.4.1.1) read a maximum current of 15ma."

Paragraph 4.5.2.5.4(b): Delete in its entirety.

Paragraph 4.5.2.5.6, Table II: Delete"4.7 ± 5%"from line 6 and substitute "2.5 ± 20%". Delete line 7.

* Paragraph 4.5.2.5.8.6: Change "0.5 volt" to "0.75 volts".

Paragraph 4.5.2.6: Add new paragraph.

"4.5.2.6 Terminal Board TB-6: 10531591.- Ground terminal 9 for the following tests:

(a) Apply -10 vdc through a 1K ohm resistor to each terminal (1 through 8) in consecutive order. The measured voltage at each terminal to ground shall be 1 VDC or less.

(b) Apply +5 VDC through 10K ohm ± 1% resistor to each terminal as specified in 4.5.2.6(a). The measured voltage across the 10K resistor shall not exceed .25 VDC.

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(c) Apply +5 VDC through 10K ohm ± 1% resistor to junction of CR9 and R9. The measured voltage across the 10K resistor shall not exceed .05 VDC.

(d) Repeat the above tests at the low temperature conditions specified in 3.5.3 and 4.5.1.4. The measured voltage specified in 4.5.2.6(a) shall be 1.5 VDC.

(e) Repeat the above tests at the high temperature conditions specified in 3.5.3 and 4.5.1.4. The measured voltage specified in 4.5.2.6(b) shall not exceed .55 VDC. The measured voltage specified in 4.6.2.6(c) shall not exceed .35 VDC."

Paragraph 5.1: Delete and substitute:

"5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with the applicable packaging data sheet listed in Section 2. The level of protection shall be as specified in the procurement document."

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